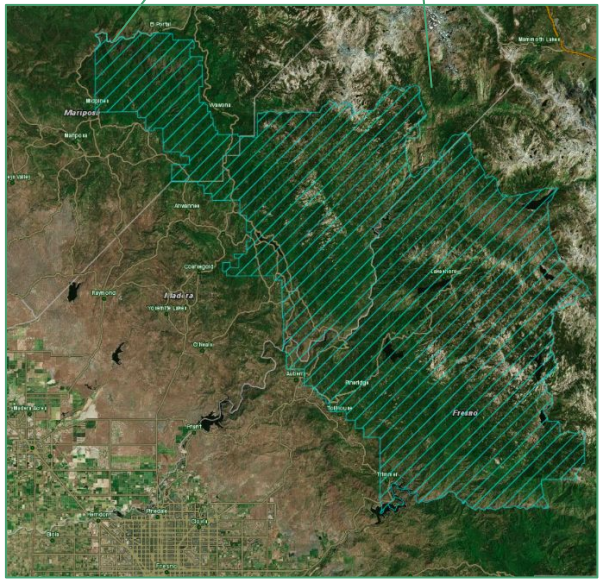
NDVI vs. soil properties in Sierra National Forest

Lauren Sharwood

Sierra National Forest (CA750)

- Parent material: Phases of granite, alluvium, and glacial deposits
- Distribution of species largely determined by altitude
- Elevations range: 1,000ft in the foothills to 10,800ft on glaciated ridges
- Mediterranean climate
- Mean annual PPT: 20 to 60 in.

Location of the Sierra National Forest Area, California

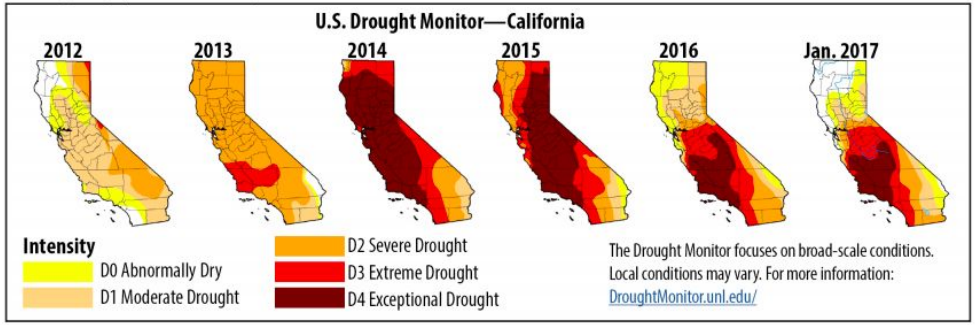


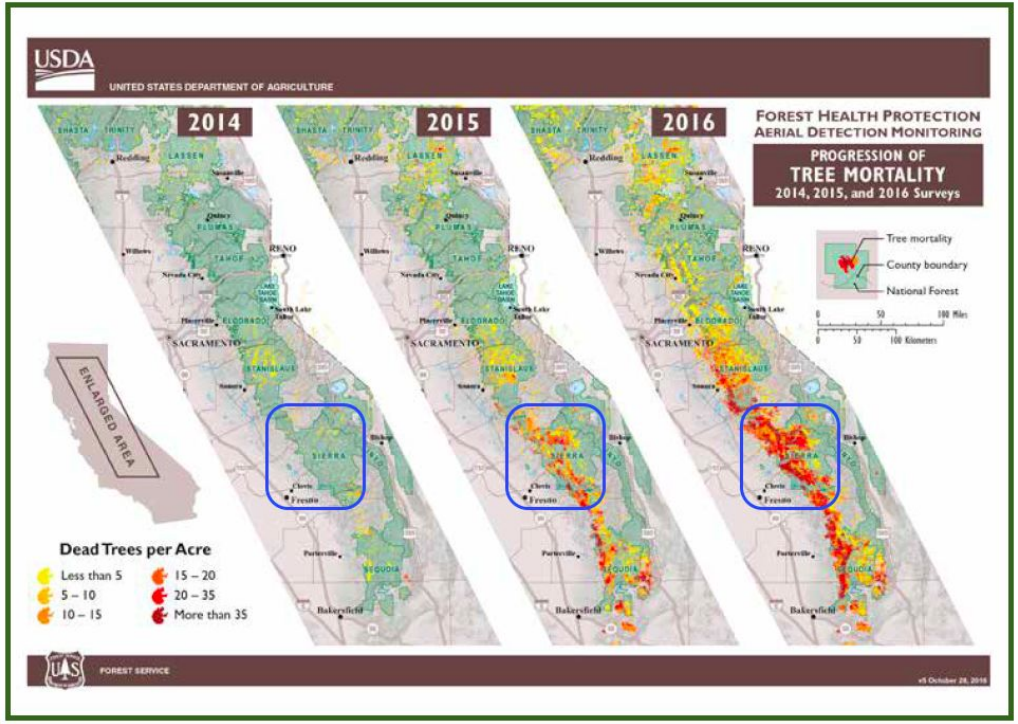
Tree Mortality

129 million trees have died in California since 2010:

- Drought → stress → foliage die back or death
- Fire suppression → overcrowding → stress → death
- Fire suppression → more intense fires → death
- Stress → Bark beetles → death

USFS: Aerial Detection Monitoring



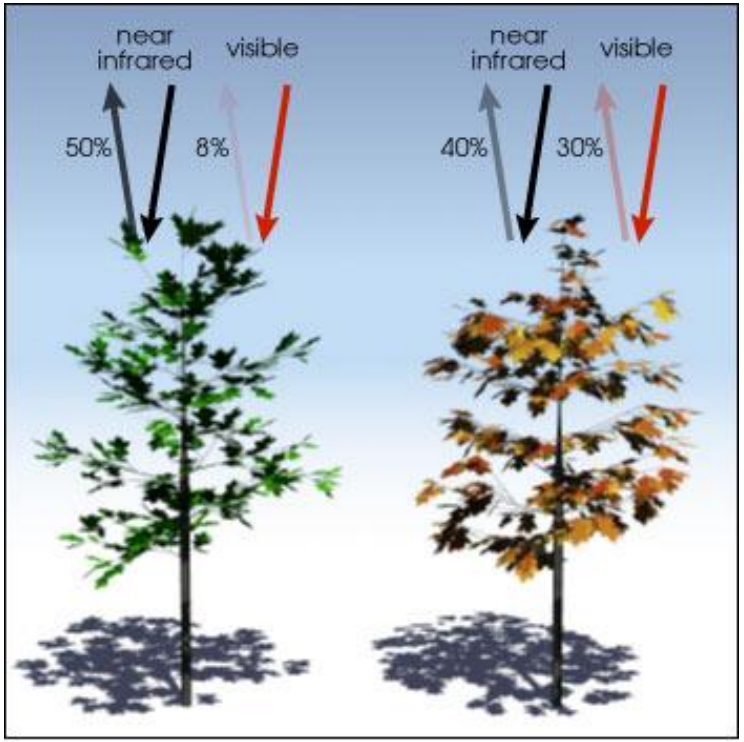


Normalized Difference Vegetation Index (NDVI)

- Proxy for vegetation health; biomass; productivity

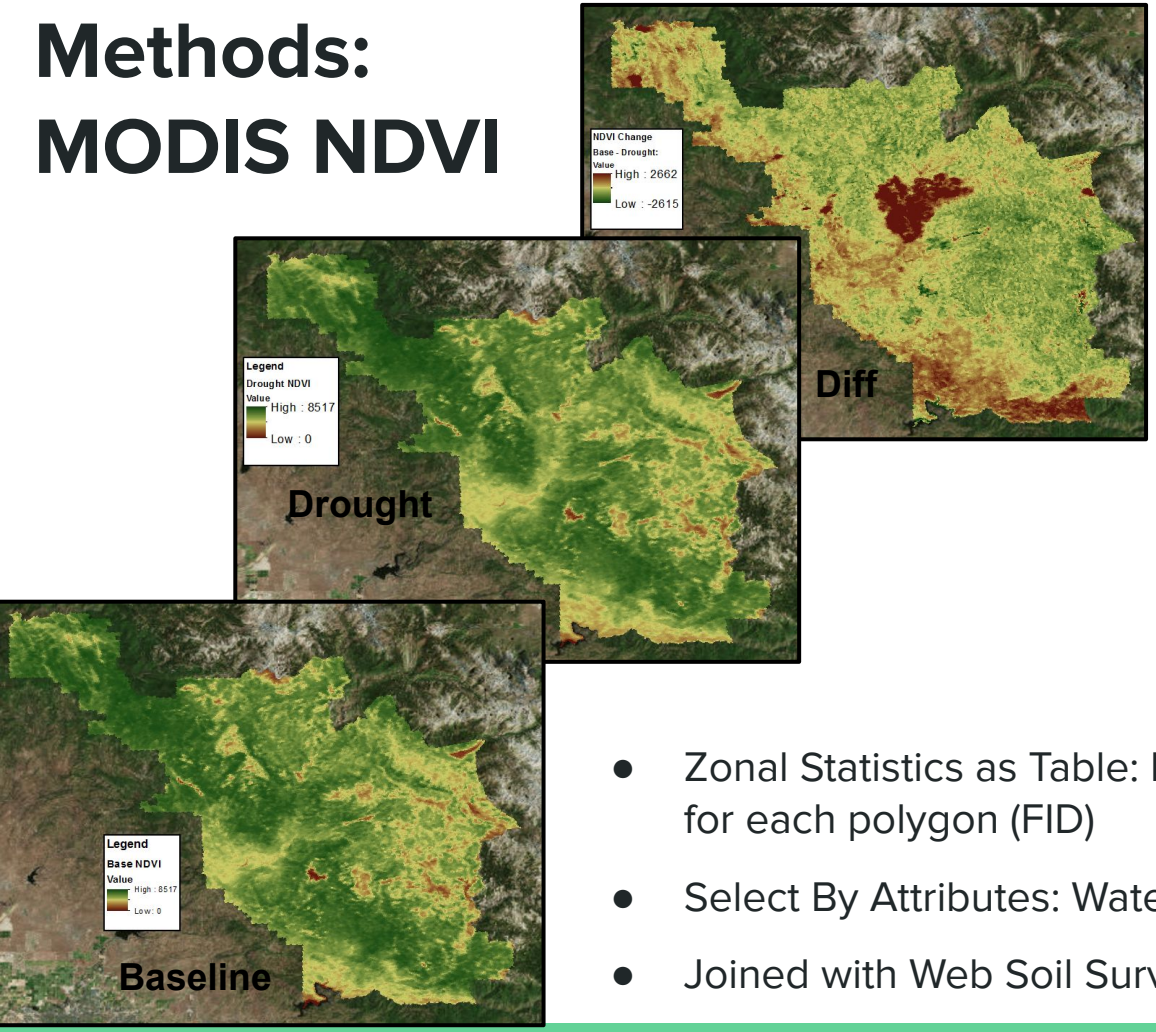
- Indicator for predicting tree and determining mortality

$$NDVI = \frac{NIR - VIS}{NIR + VIS}$$



$$\frac{(0.50 - 0.08)}{(0.50 + 0.08)} = 0.72 \qquad \frac{(0.4 - 0.30)}{(0.4 + 0.30)} = 0.14$$

Methods: MODIS NDVI



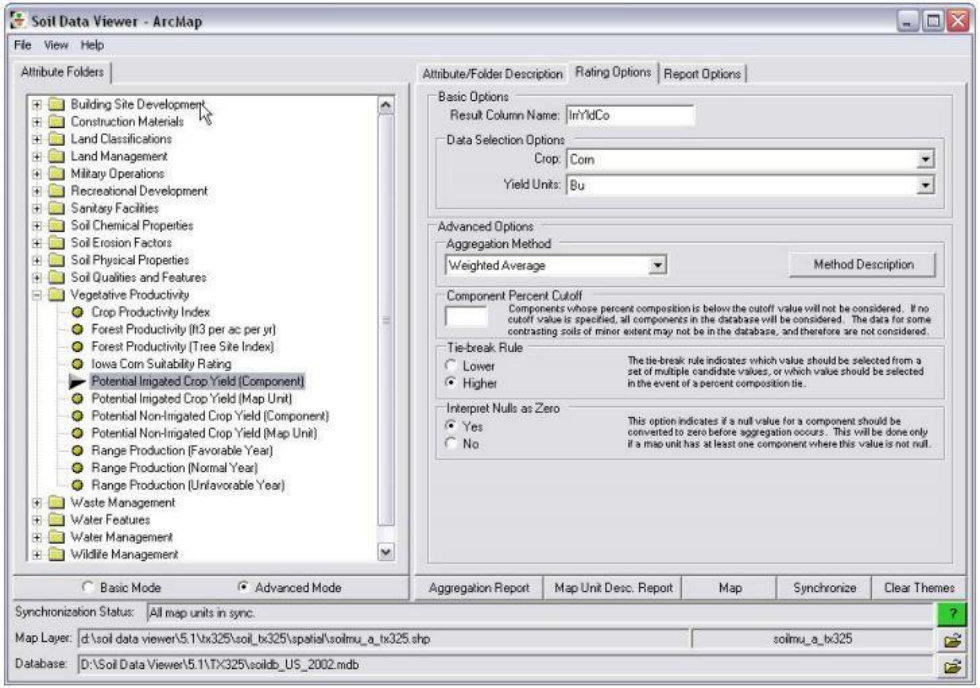
- eMODIS NDVI images downloaded from earthexplorer
- 2006-2016: July & August
- Clip to SNF (Ca750) extent
- 8 or 9 images per year → avg. NDVI / year (11 total)
- Avg of 2006 - 2010 rasters → Baseline NDVI raster
- Avg of 2012-2016 rasters → Drought NDVI raster

- Zonal Statistics as Table: Mean NDVI (baseline and drought) for each polygon (FID)
- Select By Attributes: Water; Dams excluded
- Joined with Web Soil Survey.shp

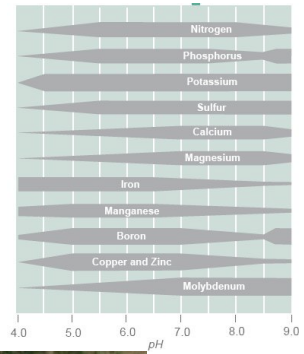
Web Soil Survey (WSS) & Soil Data Viewer

WSS Soil Properties relating to vegetation health:

- Soil Organic Matter (SOM)
- pH
- Slope
- Available Water Supply (0-25cm)
- Texture



WSS: Soil Properties

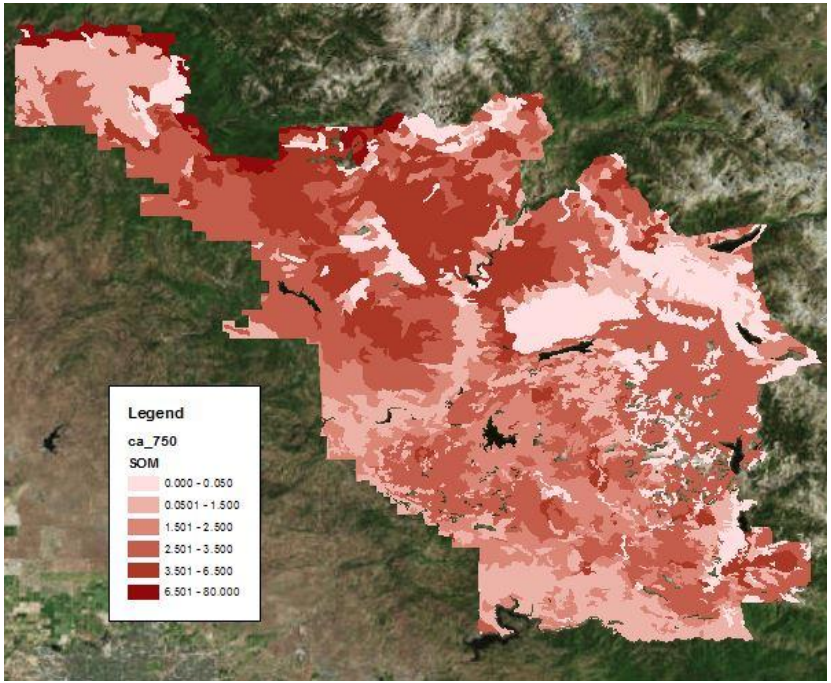


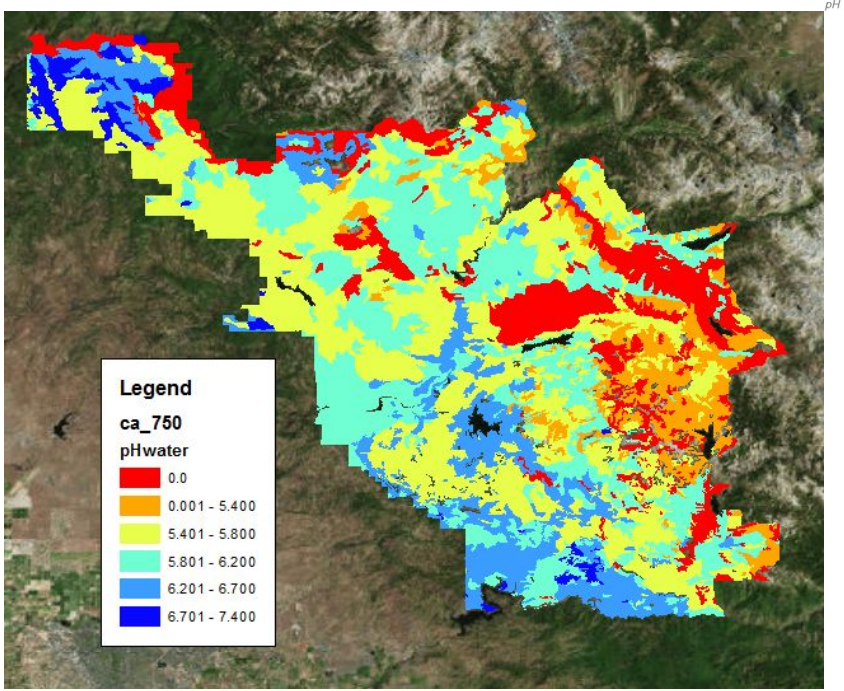
Organic Matter:

- Higher OM → Higher NDVI?

pH:

~6.5 - 7 → Higher NDVI?

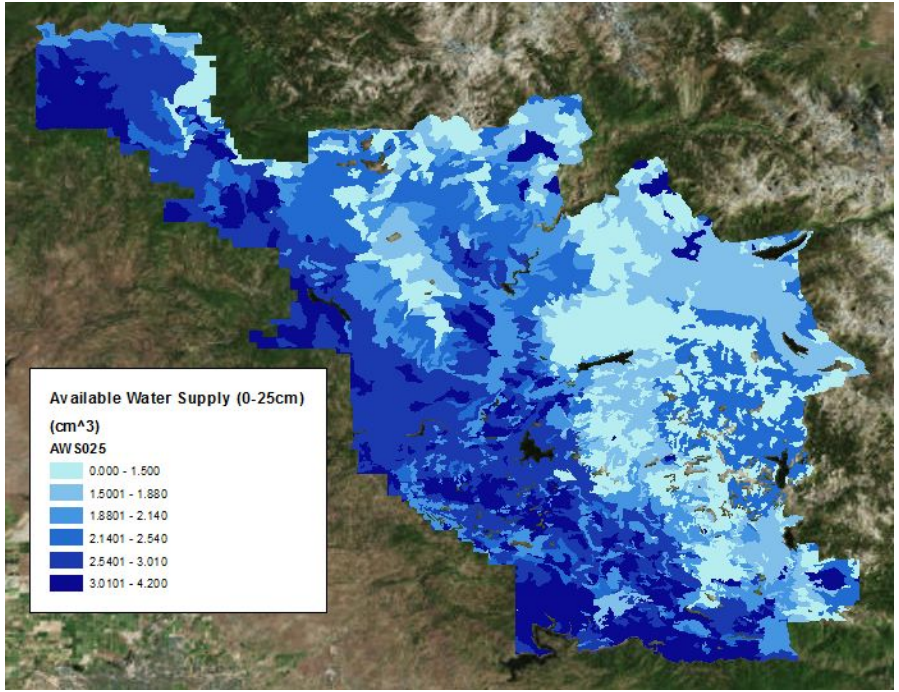




WSS: Soil Properties

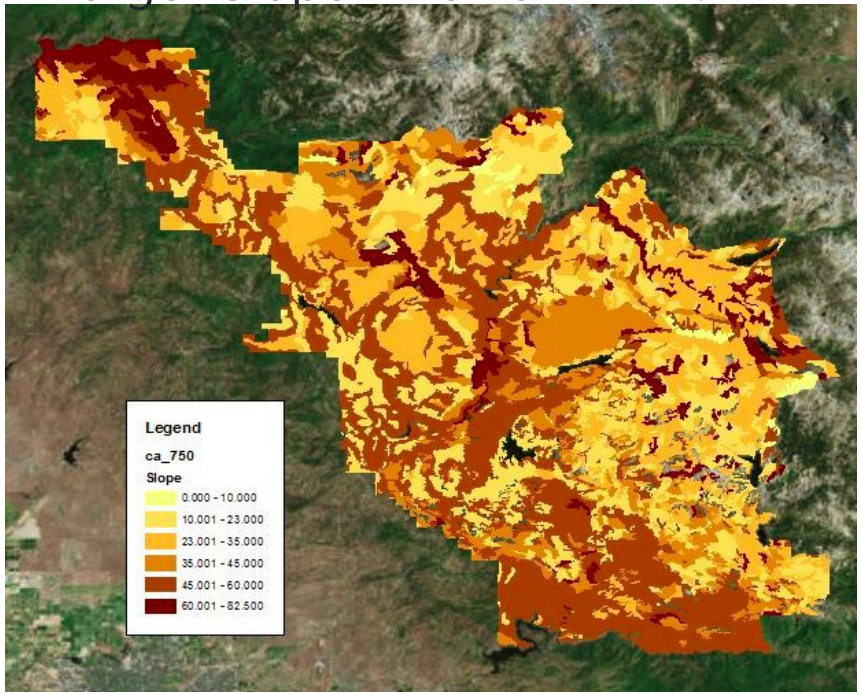
Available Water Supply:

- Greater AWS → Greater NDVI?



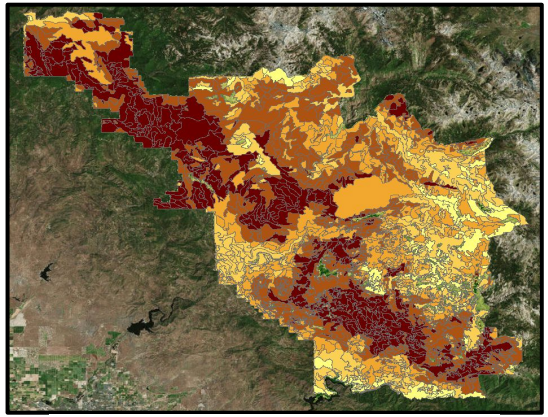
Representative Slope:

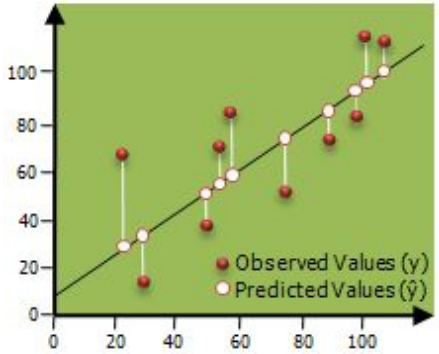
- Larger Slope → Lower NDVI?



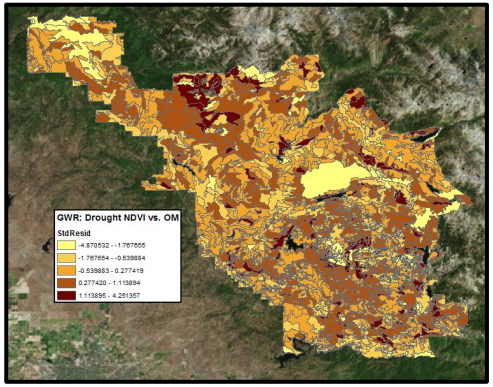
Geographically Weighted Regression (GWR)

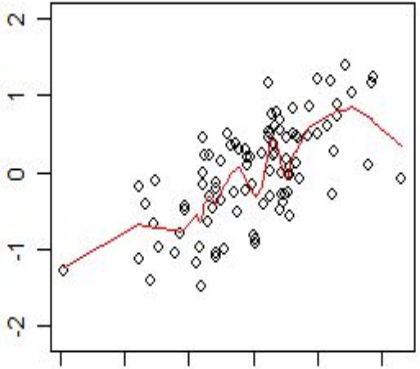
OLS Residuals:





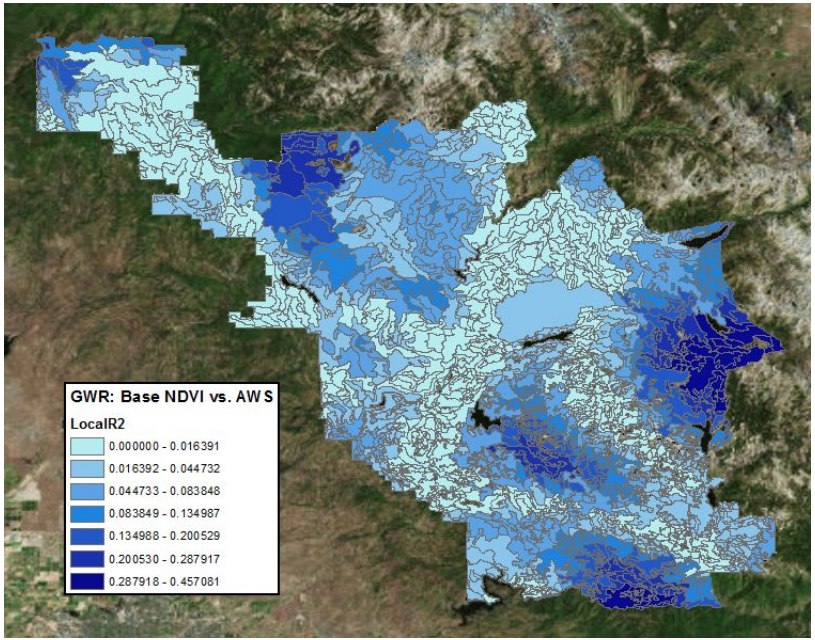
GWR Residuals:





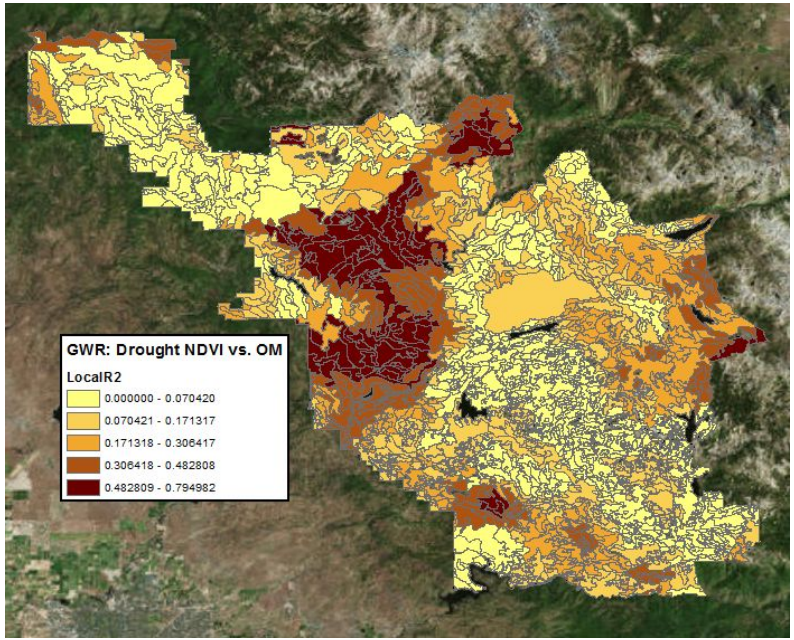
AWS (0-25cm)

VarName	Base NDVI vs. AWS	Drought NDVI vs. AWS	NDVI Diff vs. AWS
R2	0.612817516	0.5494767455	0.672062814
R2Adjusted	0.588744488	0.5303563710	0.651673355



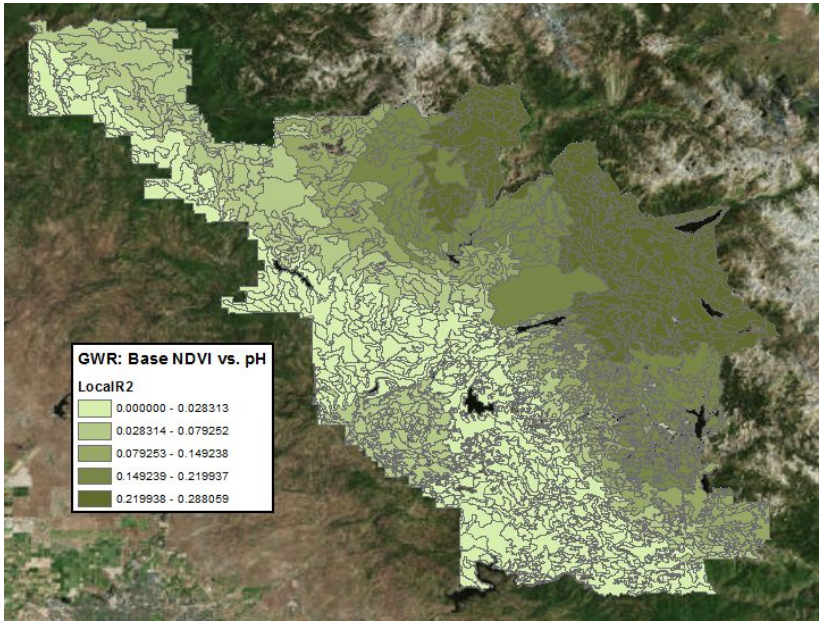
Organic Matter

VarName	Base NDVI vs. OM	Drought NDVI vs. OM	NDVI Diff vs. OM
R2	0.762873257	0.77838666	0.80620009
R2Adjusted	0.714891691	0.73354416	0.76698552



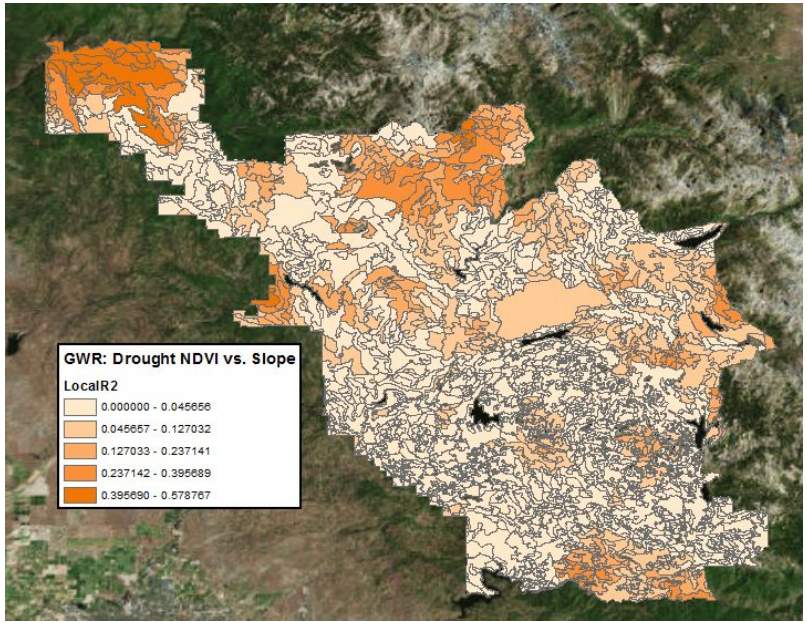
pH

VarName	Base NDVI vs. pH	Drought NDVI vs. pH	NDVI Diff vs. pH
R2	0.3816690438	0.3702724274	0.4205169219
R2Adjusted	0.3743342663	0.3604619718	0.4114892198

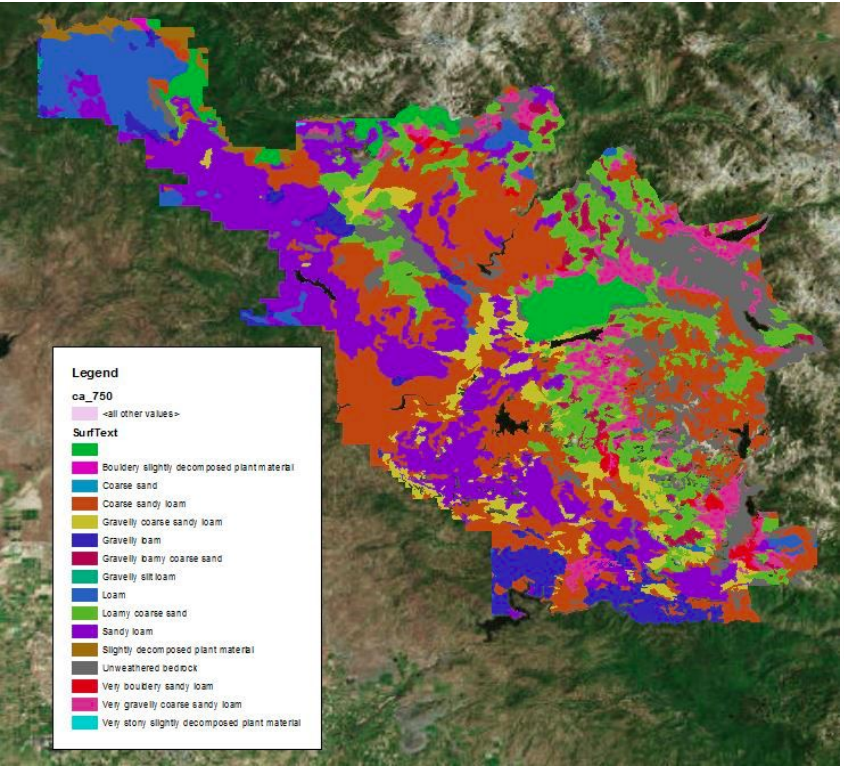


Slope

VarName	Base NDVI vs. Slope	Drought NDVI vs. Slope	NDVI Diff vs. Slope
R2	0.7472402209	0.7640004554	0.812576509
R2Adjusted	0.6907913469	0.7112946467	0.770719197



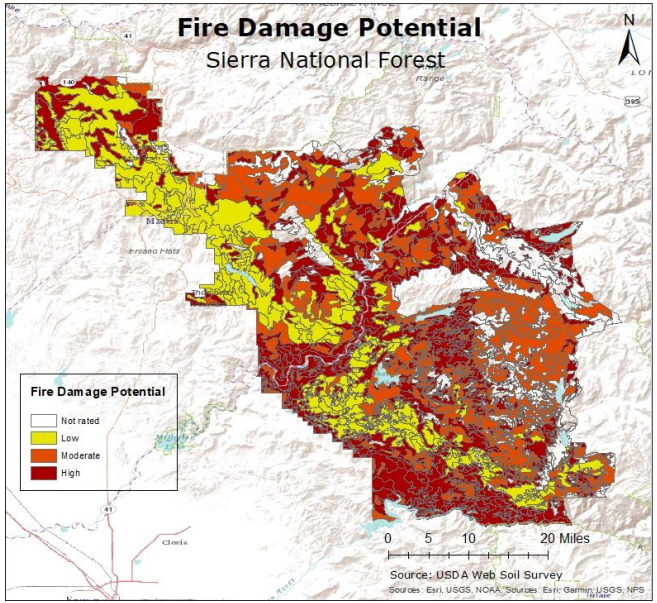
Texture



Texture:	Count	Base NDVI	Drought NDVI	Change
Very Gravely Coarse Sandy Loam	150	6123.747	6196.943	-73.196
Loamy coarse sand	299	5902.348	6059.849	-157.501
Gravelly loamy coarse sand	44	6223.231	6331.997	-108.765
Loam	55	6837.596	6646.663	190.933
Gravelly loam	57	5326.737	4938.691	388.046
Gravelly coarse sandy loam	110	6537.467	6410.553	126.913
Coarse sandy loam	473	5886.779	5764.410	122.369
Sandy loam	292	6887.982	6736.537	151.444

Limitations

- Data to include:
  - Fire
  - Bark Beetles
  - Temperature
  - PPT
- Mixed Pixels



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